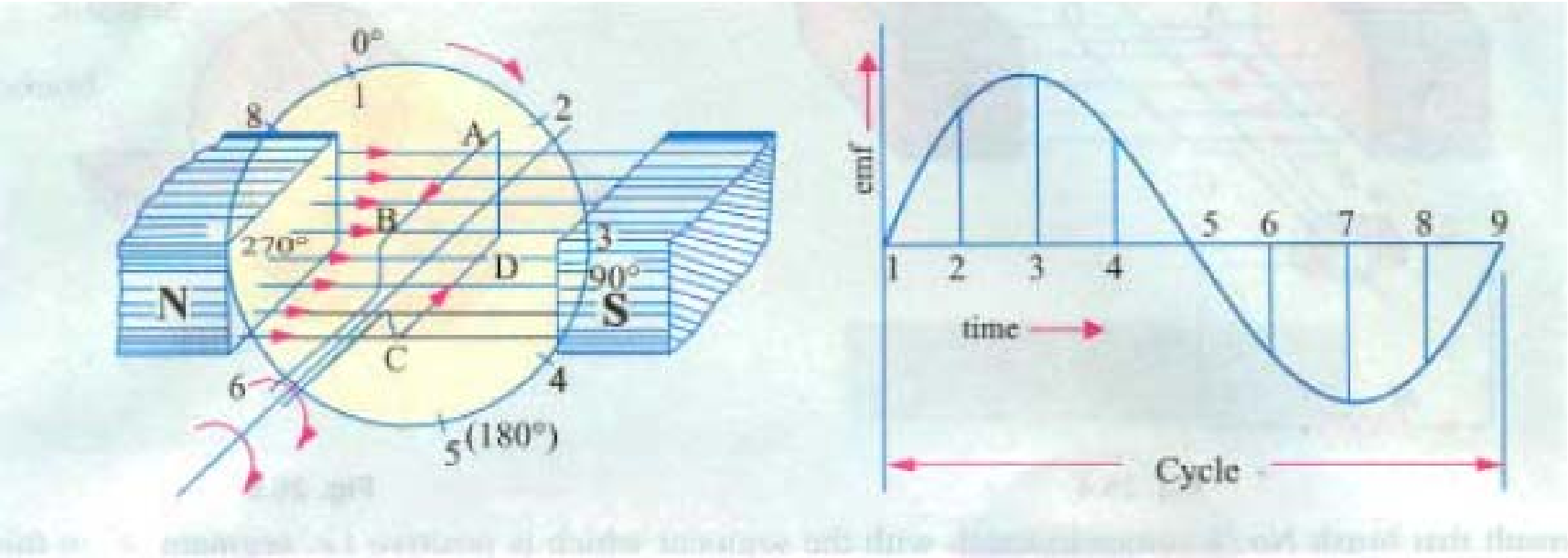
Electrical Technology

(EEA 1001)

Single Phase AC

Indian Institute of Technology
Bhubaneswar

Generation of Sinusoidal (AC) Voltage Waveform



Therefore, we find a current which we obtain from such a simple generator reverses its direction after every half revolution.

Such a current undergoing periodic reversals is known as alternating current.

Phase and Phase Difference

The phase of a vector is the angular position of the vector with respect to reference at time t=0

In Phase

Leading Phase

Lagging Phase

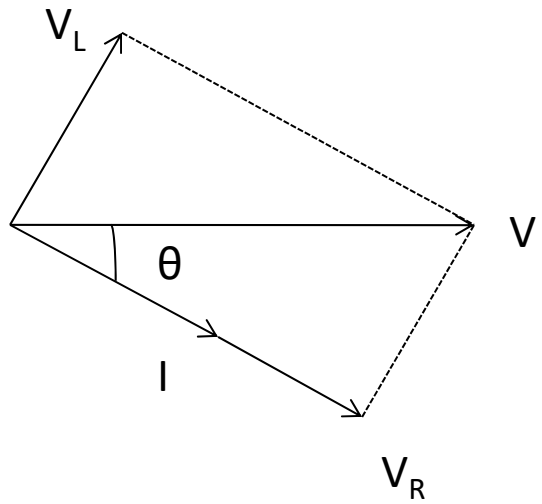
Phase difference between two or more waveform can be estimated only if frequency is same for all.

Magnitude need not same.

Phasor Diagram

If excitation source is sinusoidal then current, voltage and power will also be sinusoidal.

The diagram in which a phasor is drawn with reference to some other phasor is known as Phasor diagram.



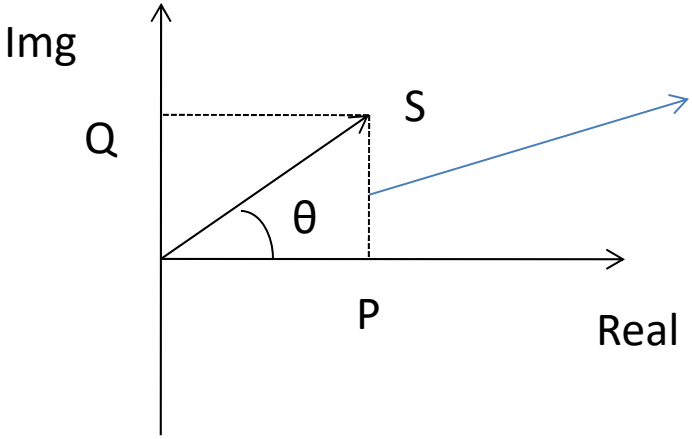
Power, Energy and Power Factor

w = instantaneous value of energy
q = instantaneous value of energy

$p = dw/dt = (dw/dq)*(dq/dt) = v*I$

Complex power $= \bar{S} = V \times I^*$

Apparent Power $= S = |\bar{S}| = VI$



Power Triangle

$P = S \cos \theta$

$Q = S \sin \theta$

Polar and Rectangular Form

$$\frac{5 - j12}{15\angle 53.1^\circ}$$

$$= \frac{13\angle -67.3^\circ}{15\angle 53.1^\circ}$$

$\angle -67.3 - 53.1$

$$= \frac{13}{15}\angle(-67.3^\circ - 53.1^\circ)$$

$$= 0.866\angle -120.4^\circ \quad = -0.438 - j0.746$$

Polar and Rectangular Form

$$(5 - j12) + 15\angle -53.1^\circ$$

$$= (5 - j12) + (9 - j11.99)$$

$$= (5 + 9) + j(-12 - 11.99)$$

$$= 14 - j23.99 \qquad = 27.77\angle -59.73^\circ$$

Polar and Rectangular Form

$$\frac{2\angle 30^\circ - 4\angle 210^\circ}{5\angle 450^\circ}$$

$$= \frac{(1.73 + j1) - (-3.464 - j2)}{5\angle 450^\circ} \qquad = \frac{(1.73 + 3.464) + j(1+2)}{5\angle 450^\circ}$$

$$= \frac{5.194 + j3}{5\angle 450^\circ} \qquad = \frac{5.99\angle 30^\circ}{5\angle 450^\circ} \qquad = \frac{5.99}{5}\angle(30^\circ - 45^\circ)$$

$$= 1.198\angle -15^\circ \qquad = 1.157 - j0.31$$

Polar and Rectangular Form

$$\left(5\angle 0^\circ + \frac{1\angle 0^\circ}{3\sqrt{2}\angle -45^\circ}\right)\times 2\angle 210^\circ$$

$$= \left(5 + j0 + \frac{1}{3\sqrt{2}}\angle -(-45^\circ)\right)\times 2\angle 210^\circ \qquad \left(\frac{1}{3\sqrt{2}}\angle 0+45^\circ\right)$$

$$= \left(5 + j0 + 0.235\angle 45^\circ\right)\times 2\angle 210^\circ$$

$$= (5 + j0 + 0.166 + j0.166)\times 2\angle 210^\circ$$

$$= ((5 + 0.166) + j(0 + 0.166))\times 2\angle 210^\circ$$

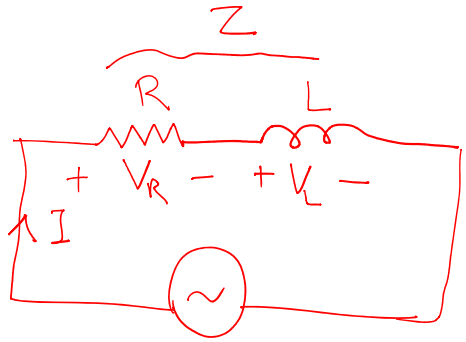
$$= (5 + j0.332)\times 2\angle 210^\circ \qquad = 5.01\angle 3.79^\circ \times 2\angle 210^\circ$$

$$= (5.01 * 2)\angle(3.79^\circ + 210^\circ)$$

$$= 10.02\angle 213.79^\circ \qquad = -8.327 - j5.572$$

Example 1

A series combination of 10 ohm resistance and 50 mH inductance is connected to a 220 V, 50 Hz supply. Estimate the current, active power, reactive power and apparent power. Also estimate the voltage across R and L and draw the phasor diagram.



220V, 50Hz

$V = 220\angle 0^\circ$

$R = 10\Omega, \quad L = 50\,mH, \quad f = 50\,Hz$

$jX_L = j\omega L = j2\pi f L = j2 \times \pi \times 50 \times 50 \times 10^{-3}$

$= j\,15.708\,\Omega$

$Z = R + jX_L = 10 + j15.708 = 18.6\angle 57.5^\circ$

$I = \frac{V}{Z} = \frac{220\angle 0^\circ}{18.6\angle 57.5} = 11.8\angle -57.5^\circ \qquad |I| = 11.8\ A.$

Power factor $= \cos\varphi = \cos 57.5 = 0.5373$ lag.

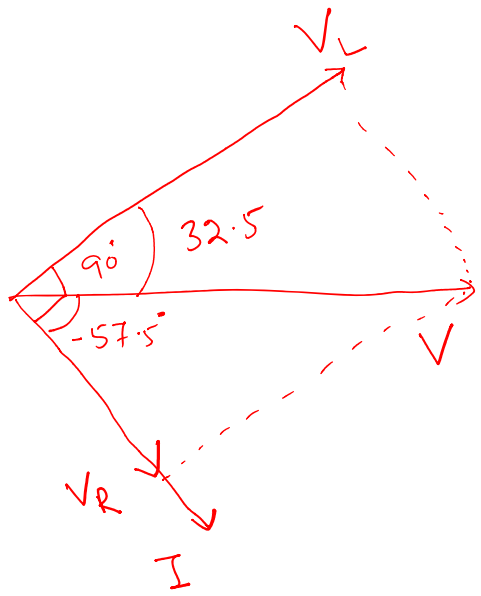
$S = VI^* = 220\angle 0^\circ \cdot 11.8\angle 57.5^\circ = 2599.2\angle 57.5 \qquad |S| = 2599.2\ VA$

$P = S\cos\varphi = 2599.2 \cos 57.5 = 1.396\ kW$

$Q = S\sin\varphi = 2599.2 \sin 57.5 = 2.192\ kVAR$

$$V_R = I \cdot R = 11.81 \angle -57.5 \cdot 10 \angle 0^\circ = 118.81 \angle -57.5$$

$$V_L = I \cdot jX_L = 11.81 \angle -57.5 \; 15.708 \angle 90^\circ = 185.58 \angle \underline{32.5^\circ}$$



$$\bar{V} = \bar{V}_R + \bar{V}_L$$

Example 2

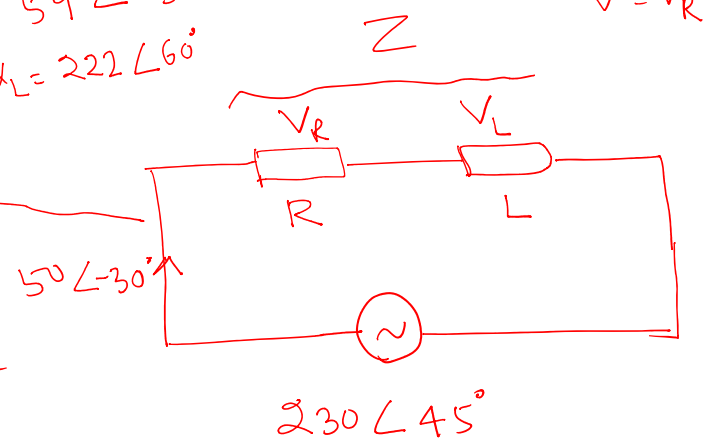
A current of 50<-30 A if flowing through a circuit consist of series connected elements, when excited by a source of 230<45 V, 50 Hz. Determine the elements of the circuit and power. Also draw the phasor diagram.

Ans: L = 14.1 mH, P = 2.9764 kW, Q = 11.1081 kVAR, S = 11.5 kVA V_R = 59.53<-30 V_L = 222.165<60

$V_R = I \cdot R = 59\angle -30^\circ$

$V_L = I \cdot jX_L = 222\angle 60^\circ$

$\bar{V} = \bar{V}_R + \bar{V}_L$



$V = 230\angle 45, \quad I = 50\angle -30^\circ$

$$Z = \frac{V}{I} = \frac{230\angle 45}{50\angle -30^\circ} = 4.6\angle 75^\circ$$

$$= 1.19 + j4.44\ \Omega$$

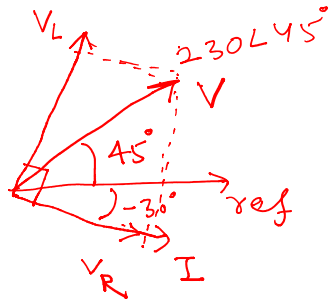
$R + jX_L - jX_C \quad = R + jX_L$

$R = 1.19\Omega, \quad X_L = 4.44\Omega \Rightarrow 2\pi f L = 4.44 \Rightarrow L = 14.1\ mH$

$S = VI^* = 230\angle 45^\circ \times 50\angle 30 = 1150\angle 75^\circ \qquad |S| = 1150\ VA = 11.5\ kVA$

$\cancel{P = S\cos\varphi = 1150\cos 75} = \quad = 2976.4 + j11108\ VA.$

$= P + jQ.$



$P = 2976.4\ W, \quad Q = 11108\ VAR$

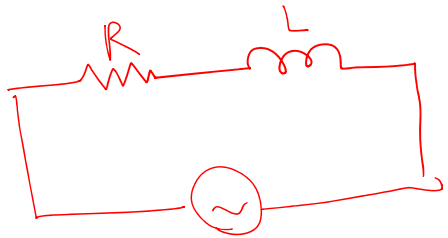
$= 2.97\ kW, \quad = 11.1\ kVAR$

Example 3

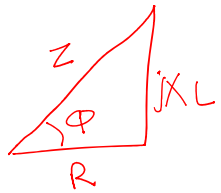
Consider a RL series circuit with an impedance angle of 50^0 at a frequency of 60 Hz. At what frequency the magnitude of the impedance will be twice the magnitude of the impedance at 60 Hz.

f = 148.332 Hz

$f_1 = 60\ Hz,\ \omega_1 = 2\pi f_1 = 376.99\ rad/sec$



Let Impedance $= Z_1 = R + j\omega_1 L = Z_1 \angle \varphi_1$



$\varphi_1 = 50^\circ$

$\Rightarrow \tan^{-1}\frac{\omega_1 L}{R} = 50^\circ \Rightarrow \frac{\omega_1 L}{R} = \tan 50 = 1.19$ — (1).

Let $f_2 \rightarrow \omega_2$ $\qquad Z_2 = R + j\omega_2 L = Z_2 \angle \varphi_2$

$Z_2 = 2Z_1 \Rightarrow \sqrt{R^2 + \omega_2^2 L^2} = 2 \times \sqrt{R^2 + \omega_1^2 L^2}$

$\Rightarrow R^2 + \omega_2^2 L^2 = 4(R^2 + \omega_1^2 L^2)$

$\frac{\omega_2}{\omega_1} = 2.47 \Rightarrow \omega_2 = 931\ rad/sec$ $\qquad f_2 = \frac{\omega_2}{2\pi} = 148\ Hz.$

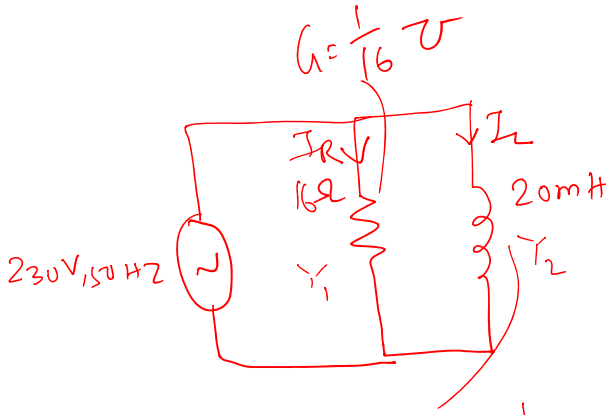
Example 4

A resistance of 16 ohm is connected in parallel to an inductance of 20 mH and the parallel combination is connected to an ac supply of 230 V, 50 Hz. Determine the current through the elements and power delivered by the source. Draw the phasor diagram.

Ir = 14.375 <0, I_L = 36.616<-90, P = 3.3008 kW

$R = 16\Omega$, $L = 20\,mH$

$V = 230V$, 50 Hz $\rightarrow 230\angle 0^\circ$



$Y_1 = G = \frac{1}{16} = 0.0625\,℧$

Inductive Susceptance $= -\frac{j}{\omega L} = -j\frac{1}{2\pi f L} = -j0.1592\,℧$

$Y_2 = -jB_L$

$Y = Y_1 + Y_2 = 0.0625 - j0.1592$

$= 0.171\angle -68.6^\circ\,℧$

$-jB_L = -j\frac{1}{\omega L}$

$I = V.Y = 230\angle 0^\circ . 0.171\angle -68.6^\circ = 39.33\angle -68.6\,A.$

$|I| = 39.33A.$

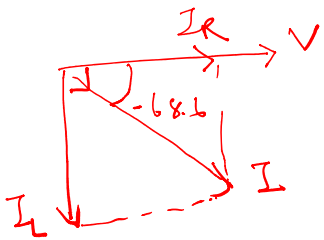
$I_R = V.G = 230\angle 0^\circ . 0.0625\angle 0^\circ = 14.37\angle 0^\circ$

$I_L = V.(-jB_L) = 230\angle 0^\circ . 0.1592\angle -90^\circ = 36.6\angle -90^\circ$

$\cos\varphi = \cos 68.6 = 0.3646$ lag

$P = VI\cos\varphi = 230 \times 39.33 \times 0.3646$

$= 3.3\,kW.$



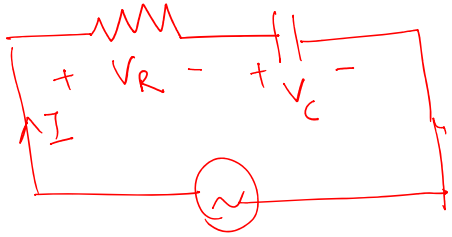
Example 5

A series combination of 12 ohm resistance and 600 uF capacitance is connected to a 220 V, 50 Hz. Estimate the current, active power, reactive power and apparent power. Also estimate the voltage across R & C and draw the phasor diagram

I = 16.7678<23.9, S = 3.6889 kVA, P = 3.3728 kW, Q = 1.4945 kVAR, V_R = 201.2136<23.9, V_C = 88.9565<-66.1

$Z = R - jX_C$

$-jX_C = \frac{-j}{2\pi f C} = \frac{-j}{2\pi \times 50 \times 600 \times 10^{-6}} = -j5.3\,\Omega$



$Z = 12 - j5.3\,\Omega = 13.12\angle -23.9^\circ$

$I = \frac{V}{Z} = \frac{220\angle 0^\circ}{13.12\angle -23.9} = 16.76\angle 23.9^\circ \quad |I| = 16.76\,A$

$S = VI^* = 220\angle 0^\circ \cdot 16.76\angle -23.9$

$= 3688.9\angle -23.9$

$= 3372.8 - j1494.5$

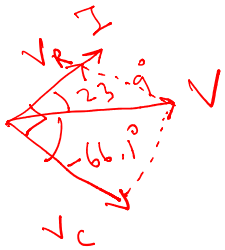
$P = 3.37\,kW, \quad Q = 1.4945\,VAR$ (Capacitive)

$= -1.4945\,VAR$

$V_R = IR = 201.2\angle 23.9$

$V_C = I(-jX_C) = 88.9\angle -66.1$

$V = V_R + V_C$



$S = 3.688\,kVA$

Example 6

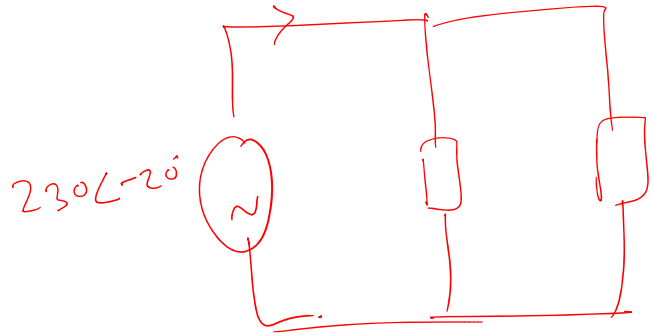
A current of 60<25 A is flowing through a circuit consits of parallel connected elements, when excited by a source of 230<-20, 50 Hz. Determine the elements of the circuit, active power and reactive power. Also calculate the current through the elements and draw the phasor diagram.

C = 0.58728 mF, S = 13800<-45, P = 97581W, Q = 97581VAR, Ir = 42.435<-20, Ic = 42.435<70

$I = 60\angle 25^\circ$, $V = 230\angle -20^\circ$, $f = 50\,Hz$

$$Y = \frac{I}{V} = \frac{60\angle 25^\circ}{230\angle -20^\circ} = 0.26\angle 45^\circ$$

$$= 0.1845 + j0.1845$$



$G = 0.1845$, $B_c = 0.1845$

$$= \omega C = 2\pi f C = 0.1845 \Rightarrow C = 0.587\ mF$$

$$I_R = V \cdot G = 42.43\angle -20^\circ$$

$$I_c = V \cdot (jB_c) = 42.43\angle 70^\circ$$

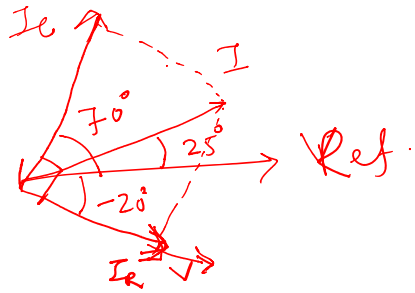
$$S = VI^* =$$

$S = 13.8\ kVA$

$P = 9.75\ kW$

$Q = 9.75\ kVAR$ (Capacitive)

$$\bar{I} = \bar{I}_R + \bar{I}_c$$

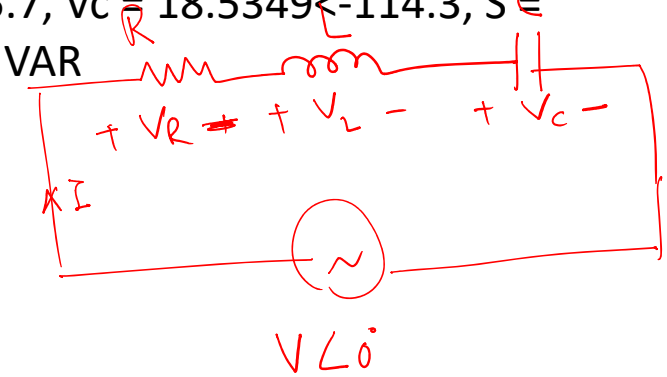


Example 7

A series circuit consists of R = 75 ohm, L = 125 mH, & C = 200 uF. The circuit is excited by a sinusoidal source of value115 V, 60 Hz. Determine the voltage across various elements. Calculate the current and power. Draw the phasor diagram.

$R = 75\,\Omega$, $L = 125$ mH, $C = 200\ \mu F$.

I = 1.397<-24.3, Vr = 104.8125<-24.3, Vl = 65.8557<65.7, Vc = 18.5349<-114.3, S = 160.7125 VA, Phi = 24.3, P = 146.4739W, Q = 66.1355 VAR



$Z = R + j(X_L - X_C)$

$X_L = 2\pi f L = 47.1\,\Omega$

$X_C = \frac{1}{2\pi f C} = 13.27$

$Z = 75 + j(47.1 - 13.27) = 75 + j33.8$

$= 82.28\angle 24.3^\circ$

$I = \frac{V}{Z} = \frac{115\angle 0^\circ}{82.28\angle 24.3^\circ} = 1.39\angle -24.3^\circ$

$V_R = 104.8\angle -24.3 = IR$

$V_L = 65.8\angle 65.7^\circ = I(jX_L)$

$V_C = 18.53\angle -114.3^\circ = I(-jX_C)$

$\bar{V} = \bar{V}_R + \overbrace{\bar{V}_L + \bar{V}_C}^{V_X}$

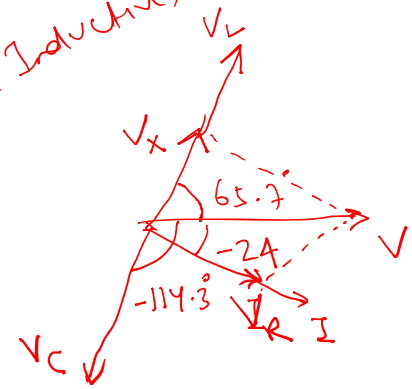
$= \bar{V}_R + \bar{V}_X$

$S = VI^*$

$|S| = 160$ VA.

$P = 146$ W

$Q = 66$ VAR (Inductive)



Example 8

A RLC parallel circuit consists of R = 50 ohm, L = 150 mH, & C = 100 uF. The circuit is excited by a current source of 5<0 A, 100 Hz. Calculate the voltage & current in the various elements. Determine the apparent, active & reactive power delivered by te source. Draw the phasor diagram.

V = 89.4454,-69, Ir = 1.7889<-69, Il = .9481<-159, Ic = 5.6172<21, S = 447.227 VA, P = 160.2718 W, Q = -417.5224 VAR.

Example 9

A load absorbs 2.5 kW at a power factor of 0.707 lagging from a 230 V, 50 Hz source. A capacitor is connected in parallel to the load in order to improve the power factor to 0.9 lag. Determine the value of capacitor.

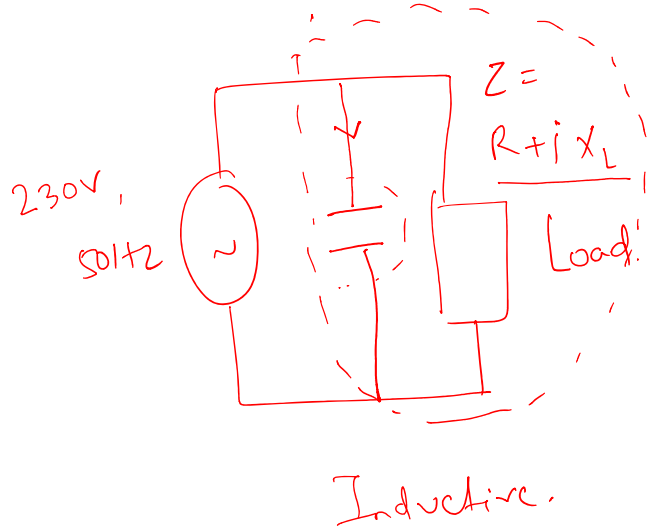
C = 77.6815 uF.

P = 2.5 kW, P.F = .707, V = 230 V.

$\cos\varphi_1 = 0.707$

$S_1 = \frac{P}{\cos\varphi_1} = \frac{2.5\times10^3}{0.707} = 3.53$ kVA

$Q_1 = S_1 \sin\varphi_1 = 3.53\times10^3 \times \sin 45^\circ = 2.5$ kVAR



Case - II

$\cos\varphi_2 = 0.9$

$S_2 = \frac{P}{\cos\varphi_2} = \frac{2.5}{0.9} = 2.77$ kVA

$Q_2 = 1.207$ kVAR

$Q_C = 2.5 - 1.207$ 1291 VAR. $= V I_C$

$I_C = \frac{1291}{V} = 5.631$ A.

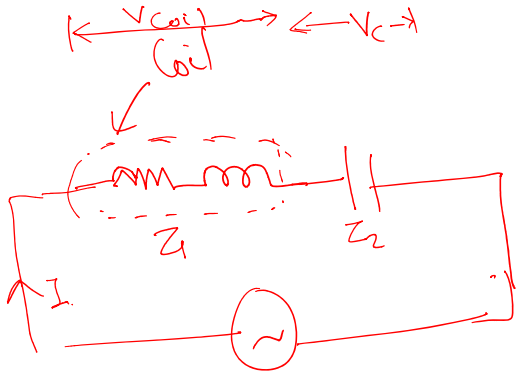
$X_C = \frac{V}{I_C} = \frac{230}{5.631} = 40.97\ \Omega$.

$C = 77.68$ μF.

Example 10

An inductive coil of power factor 0.8 lagging is connected in series with a 120 uf capacitor. When the series circuit is connected to a source of frequency 50 Hz, it was observed that the magnitude of voltage across the coil and capacitor are equal. Determine the parameters of the coil.

R = 21.2123 ohm, L = 50.7 mH.



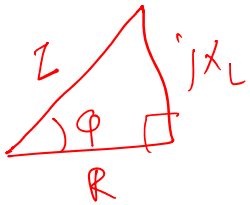
$Z_1 = R + jX_L = Z_1 \angle \varphi$.

$V_{(coil)} = I Z_1$

$V_C = I X_C$

$I Z_1 = I X_C$

$\Rightarrow Z_1 = X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 120 \times 10^{-6}} = 26.52$ 50 Hz.



$Z_1 = 26.52\,\Omega$, p.f = 0.8

$R = Z\cos\varphi = 21.21\,\Omega$

$X_L = Z\sin\varphi = 15.92\,\Omega$

$\downarrow$

$L = 50.7$ mH.

Example 11

Three impedances 12-j0 ohm, 5+j8 ohm, 0-j7 ohm are connected in parallel. This parallel combination is connected in series with an impedance of 4+j6 ohm across a 230 V source. Determine the current through each impedance and power.

I1 = 11.7996<-40.2, I2 = 15.009<-98.2, I3 = 20.2278<49.8, I4 = 21.1304<-19.4

S1=1670.8VA, P1=1670.8W, S2=2125.2<58, P2=1126.2W, Q2= 1802.3, S3=2864<-90, P3=0,Q3=-2864